

Network Mathematics Now

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Network Mathematics

Network Mathematics is a research project that aims to uncover the *structural relationships underlying mathematics* by extracting information from mathematical texts and organizing it as networks.

Core idea

- Mathematical knowledge is distributed across papers, books, and online resources.
- These texts implicitly encode a large graph of relationships between definitions, theorems, concepts, examples and explanations.
- By using NLP and knowledge extraction tools, we can transform this implicit structure into explicit **knowledge graphs**.

Goal

Build computational representations of mathematical knowledge that support search, discovery, and automated reasoning.

Motivation

Modern mathematics is extremely large, deep and fragmented.

Challenges

- Millions of papers across thousands of journals, no domain experts in all
- Mathematical concepts reused across many areas, subtle differences
- Dependencies between results are often implicit, even to authors

Opportunity

- Natural language processing and knowledge graph technology
- Open mathematical corpora (arXiv, textbooks, wikis)
- Structured mathematical databases (OEIS, LMFDB, Wikidata, code4math...)

Vision

Represent mathematics as a *network of concepts* linking definitions, theorems, proofs, examples, and applications.

Methodology

Network Mathematics combines tools from NLP, knowledge representation, and category-theoretic perspectives on mathematical structure.

Pipeline

- ① Collect mathematical corpora (papers, textbooks, wikis)
- ② Extract mathematical entities
 - definitions
 - theorems
 - mathematical objects
- ③ Identify relationships between them
 - dependency
 - usage
 - analogy
- ④ Build knowledge graphs representing the conceptual structure

The resulting graphs provide a **computational map of mathematics**.

Subprojects and Resources

Several subprojects contribute to the Network Mathematics ecosystem.

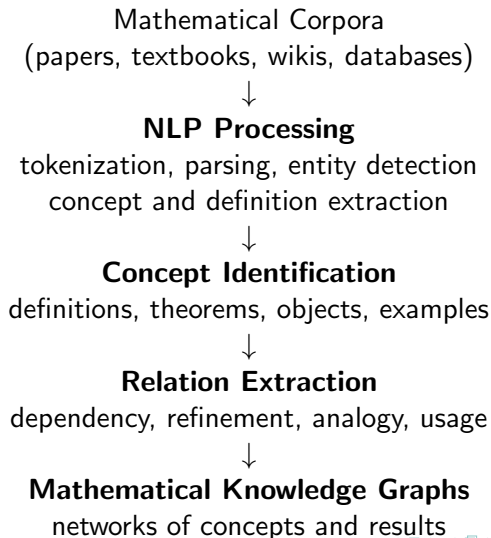
Examples

- **MathGloss**: automatic construction of mathematical glossaries
- **Concept extraction**: identifying definitions and key terms in text
- **Argument structure**: mapping how mathematical reasoning unfolds
- **Mathematical corpora**: datasets derived from textbooks and journals

Long-term vision

- A large-scale **knowledge graph of mathematics**
- Tools for exploring conceptual dependencies
- Support for AI-assisted mathematical discovery

Architecture of Network Mathematics



Foundation: Maths has underlying structural organization

Category Theory

- Provides a language for describing math structures and relations.
- Math concepts as objects, dependencies/transformations as morphisms.
- Knowledge graphs extracted from texts interpreted as *approximations*

Logic and Type Theory

- Mathematical texts encode arguments and proofs.
- Through the Curry–Howard correspondence:

proofs $\leftrightarrow$ programs $\leftrightarrow$ constructions

- Extracting argument structure from text reveals computational and logical content of mathematics.

Goal: Bridge *textual mathematics* and *structural mathematics* through computational representations.